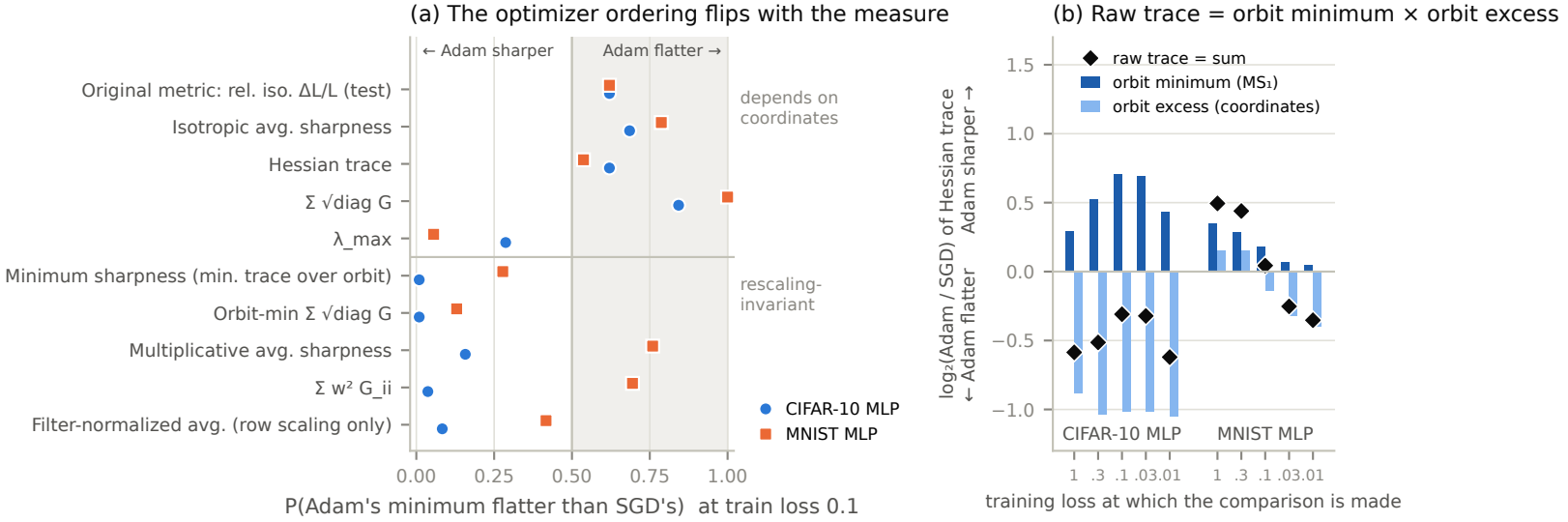
(a) The optimizer ordering flips with the measure
← Adam sharper
Adam flatter →
Original metric: rel. iso. ΔL/L (test)
Isotropic avg. sharpness
Hessian trace
Σ √diag G
λ_max
Minimum sharpness (min. trace over orbit)
Orbit-min Σ √diag G
Multiplicative avg. sharpness
Σ w² G_ii
Filter-normalized avg. (row scaling only)
depends on coordinates
rescaling-invariant
CIFAR-10 MLP
MNIST MLP
0.00
0.25
0.50
0.75
1.00
P(Adam's minimum flatter than SGD's) at train loss 0.1
(b) Raw trace = orbit minimum × orbit excess
raw trace = sum
orbit minimum (MS₁)
orbit excess (coordinates)
log₂(Adam / SGD) of Hessian trace
← Adam flatter Adam sharper →
1.5
1.0
0.5
0.0
−0.5
−1.0
CIFAR-10 MLP
MNIST MLP
1 .3 .1 .03 .01
1 .3 .1 .03 .01
training loss at which the comparison is made